Singapore School Kinderland Secondary-2

Computer Science (0478 / 2210)

Term 2 • Academic Session 2026-27

Class Test-2: Error Detection & Cryptographic Security (TB Pg 54–74) [ANSWER KEY]

Date: _____ **Student Name:** TEACHER ANSWER KEY & MARK SCHEME

Total Marks: 25 Marks **Marks Obtained:** 25 / 25

Teacher's Signature: _____ **Parent's Signature:** _____

Section A: Fill in the Blanks using the Security Word Bank [5 Marks]

Word Bank: **Parity Block • Check Digit • Checksum • Asymmetric • Automatic Repeat Request (ARQ)**

1. A _____ combines horizontal and vertical parity checks to locate and automatically correct a flipped bit. **[Parity Block]**

2. A _____ is calculated on identification numbers (barcodes) to detect human transcription and transposition errors. **[Check Digit]**

3. In _____ encryption, two mathematically paired keys (a Public Key and a Private Key) are utilized. **[Asymmetric]**

4. The _____ algorithm computes a numerical value from a block of data at sender and recalculates it at destination. **[Checksum]**

5. _____ is a transmission protocol that uses positive acknowledgements (ACK) and timeout timers to trigger retransmission. **[Automatic Repeat Request (ARQ)]**

Section B: Write [T] for True or [F] for False [5 Marks]

No.	Technical Statement	Answer
1.	If two bits flip in the same byte during transmission, a standard single parity check will fail to detect the error.	[T]
2.	In symmetric encryption, the sender and receiver use completely different, non-shared cryptographic keys.	[F]
3.	An Echo Check requires double the network bandwidth because the receiving computer sends all data back to the sender.	[T]
4.	Digital signatures are encrypted using the sender's private key to provide authenticity and prove non-repudiation.	[T]
5.	In an ARQ protocol, if a timeout occurs without receiving an ACK, the sender terminates transmission permanently.	[F]

Section C: Match Cryptographic Concepts with Security Roles [4 Marks]

Security Concept	Role & Mechanism	Answer
1. Public Key	A. Kept secret by owner; used to decrypt confidential messages	1 → C

Section D: Multiple Choice Questions (Cambridge Exam Pattern) [4 Marks]

1. What is the fundamental security flaw associated with Symmetric Encryption?

(A) The shared secret key must be transmitted across the network, risking interception

(B) It can only encrypt numbers (C) Encryption speed is too slow

2. If a customer mistypes an account number as "4589" instead of "4598", which error type is this?

(A) Transposition Error (B) Transcription Error (C) Parity Error

3. In an ARQ protocol, what packet does the receiver transmit if it detects a checksum mismatch?

(A) Negative Acknowledgement (NAK) (B) Positive ACK (C) Timeout

4. When user A sends an encrypted message to user B, which key is used to encrypt the plain text?

(A) User B's Public Key (B) User A's Private Key (C) User A's Public Key

Section E: Parity Block Matrix Analysis & ARQ Protocol Mechanics [7 Marks]

1. Parity Block Matrix Error Pinpointing Problem: (4 Marks)

A transmission uses **Odd Parity**. The received block of bytes and its parity byte are shown below:

Byte	b1	b2	b3	b4	b5	b6	b7	Parity Bit
Byte 1	1	0	0	1	0	1	0	0
Byte 2	0	1	1	0	1	0	1	1
Byte 3	1	1	0	**1**	1	1	0	0
Byte 4	0	0	1	1	0	0	1	0
Parity Byte	1	1	1	**0**	1	1	1	0

a) Identify which byte row contains the corrupted parity count: **Byte 3 (row has 5 ones + parity 0 = 5 ones, wait: 1+1+0+1+1+1+0 = 5 ones, parity 0 = 5 is odd; wait: column 4 has 1+0+1+1 = 3 ones + parity 0 = 3 ones; error is at Byte 3, Bit 4!) [1.5 marks]**

b) Identify which bit column has flipped: **Bit 4 [1.5 marks]**

c) State the corrected bit value: **Should be inverted from 1 to 0. [1 mark]**

2. ARQ & Encryption Scenario Question: (3 Marks)

Explain how the combination of **Timeout Timers** and **Negative Acknowledgements (NAK)** prevents infinite communication deadlocks when transmitting data over noisy satellite channels:

Model Answer: When a packet is corrupted, the receiver immediately sends a NAK triggering an immediate retransmission without waiting for timeout. If a packet or its ACK is entirely lost in transmission, the sender's Timeout Timer expires, ensuring the packet is automatically resent rather than leaving the system in an indefinite hanging state. [3 marks]

Teacher's Assessment Feedback & Evaluation Rubric:

Assessment Criteria	Max Marks	Marks Awarded	Teacher Qualitative Remarks & Feedback
1. Error Detection Methods & Check Digits	8	**8**	Strengths: **Flawless parity matrix analysis & protocol logic** Focus Area: **Practice Asymmetric key pair operational roles** Parent's Ack: _____
2. Cryptography, Hashing & SSL Certificates	8	**8**	
3. Parity Block Solving & ARQ	9	**9**	